

Lab 4: VibeCoding — Connect Four vs AI

Software Engineering Laboratory Submission — Final LLM Chat History

STUDENT REPOSITORY	TARGET FOLDER	AI CODING ASSISTANT	FINAL STATUS
abhiraamadiga/SE_submissions	Lab-4	Antigravity IDE (Gemini)	All 4 Tasks Completed & Verified

Complete Lab Tasks Summary

Task	Requirement	File	Implemented Solution
Task 1	Complete Win Detection	board.py	Added diagonal checks (1, 1) and (1, -1) to detect all 4 directions.
Task 2	Game Termination & Input	game.py	Protected against invalid moves, full columns, EOF crashes; instant win/draw halt.
Task 3	Intelligent AI	ai.py	Added immediate win detection, opponent threat blocking, and center preference.
Task 4	Move-Level Feedback	game.py	Added clear placement feedback message printed once per actual move.

Prompt and Response History

User Prompt 1

"C:\Users\abhi1\Downloads\Lab_4_VibeCoding_Student_handout (1).pdf"

this contains instructions i want u to rename 15_connect_four to Lab-4 and also wait until i tell u to make changes and add this repo as remote git remote add origin https://github.com/abhiraamadiga/SE_submissions.git

dont do any changes in the 15_connect_four
dont push or add anything

AI Assistant Response

Completed initial setup:

- Renamed directory from 15_connect_four to Lab-4 .
- Configured remote origin to https://github.com/abhiraamadiga/SE_submissions.git .
- Preserved all original starter files untouched.

User Prompt 2

i made 10-seconds video whts the nexr step?

AI Assistant Response

Recorded "before" video saved into Lab-4/videos/before.mp4 . Outlined the 4 tasks to be implemented and committed sequentially.

User Prompt 3

Complete task 1

AI Assistant Response

Task 1 Completed: Fixed diagonal win detection defect in `board.py`.

- Added diagonal directions `(1, 1)` and `(1, -1)` to `Board.winner()`.
- Verified with unit tests that all four directions detect wins and sequences `< 4` do not count.

User Prompt 4

complete task-2

AI Assistant Response

Task 2 Completed: Game termination, draw handling, and input validation in `game.py`.

- Validated integer column numbers between 1 and 7; rejected full columns without board alteration.
- Protected against unhandled `EEOFError` and `KeyboardInterrupt`.
- Terminates game immediately on a win or draw before requesting another turn.

User Prompt 5

complete Task-3

AI Assistant Response

Task 3 Completed: Intelligent decision-making in `ai.py`.

- Immediate win: Takes winning move when one exists for AI.
- Immediate block: Blocks opponent's 3-in-a-row winning threat.
- Center priority: Prioritizes central columns (`[3, 2, 4, 1, 5, 0, 6]`) and never picks a full column.

User Prompt 6

compelte task-4

AI Assistant Response

Task 4 Completed: Move-level feedback in `game.py` !

1. Implementation Details

Added feedback printed immediately after each valid disc drop in `game.py`:

```
# Task 4: Move-level feedback (occurs once per actual move)
player_label = "Player X" if self.turn == "X" else "AI (0)"
print(f"{player_label} placed a disc in column {col + 1}.")
```

2. Verification

- ✓ **Single Occurrence:** Printed exactly once per actual move.
- ✓ **Zero Side-effects:** Never triggered during AI simulation, win checks, or invalid inputs.
- ✓ **Clear attribution:** Clearly distinguishes between human player (`Player X`) and AI (`AI (0)`).

Done when criteria satisfied: Concise placement feedback occurs once per actual move. All 4 tasks are complete and verified!